

How do I determine what the Column and Row refer to in the result table?

The **Column** and **Row** are variables designed to uniquely identify each grid cell in the grid definition. In the case of the *U.S. County* grid definition, the **Column** refers to the *state FIPS code* and the **Row** refers to the *county FIPS code*. One way to get a good sense of the **Column** and **Row** variables is to create a map (discussed in the next chapter) and then view where particular **Column** and **Row** variables occur in the map.

Chapter 7

Aggregating, Pooling, and Valuing

In this chapter...

- **Get an overview of valuation, discounting, and pooling.**
- **Configure an Aggregation, Pooling, and Valuation (APV) file.**
- **Sort and pool incidence results.**
- **Learn the differences between the pooling methods.**
- **Assign economic values to incidence results.**
- **Aggregate incidence results and valuations.**
- **Run, save, and re-open an APV configuration.**
- **View and export APV results.**

This section presents an introduction to valuation, discounting, and pooling. Most BenMAP-CE users find this portion of the program the most complex to understand and use. You may find yourself referring to this chapter frequently.

Once you have created a configuration results file with incidence results based on your two air quality grids (refer to Chapter 6), you can use the Aggregate, Pool, and Value feature to combine the incidence results and place an economic value on the combined results. You have two options.

- **Create a New Configuration to Aggregate, Pool, and Value results.**

You can create a new type of configuration, termed an Aggregation, Pooling, and Valuation (APV) Configuration. This allows you to (1) specify the geographic level at which you want to report your results, (2) specify how you might want to combine or "pool" the incidence results, and (3) specify how to assign an economic value to the health incidence results. These selections can be saved in an APV Configuration file (.apvx) and used to calculate results, which are stored in an APV Results file (.apvrx).

Fundamental Concept – APVX and APVRX Files

An **Aggregation, Pooling, and Valuation (APV) Configuration** file (*.apvx) stores your user-specified BenMAP-CE preferences regarding how to aggregate your results, whether and how to pool your results, and any economic valuation functions you have applied. For example, an APV file might aggregate your estimated change in incidence to the U.S. County level, it might pool across multiple hospital admission health impact functions, and it could include an economic valuation function.

Results derived from an APV configuration will be stored along with the configuration details in an **Aggregation, Pooling, and Valuation (APV) Configuration Results** file (*.apvrx). The results in this file are the aggregated, pooled, and/or valued results for a given scenario.

- **Open Existing Configuration for Aggregation, Pooling, and Valuation.** You can load an existing APV Configuration file, edit the configuration, save it with the same or a different name, and then proceed to calculating the results.

7.1 Introduction to Valuation, Discounting, and Pooling

Valuation generally refers to placing a monetary value on estimated health incidence. In the example below, we discuss U.S. dollar values and provide a brief introduction to discounting, which has to do with placing less weight on things occurring in the future than on things occurring today. Finally, we discuss pooling, which has to do with combining comparable results.

7.1.1 Overview of Economic Valuation

Improvements in ambient air quality generally lower the risk of developing an adverse health effect by a fairly small amount across a large population. A lower risk for everyone means that fewer cases of the adverse health effect are expected, although we cannot predict which people would be spared. Therefore, the health benefits conferred on individuals by a reduction in pollution are actually reductions in the risk of having to

endure certain health problems. Monetizing the benefits of a reduction in air pollution involves estimating society's willingness to pay (WTP) for these reductions in risk, or the observed Cost of Illness (COI) for an effect, and is typically referred to as valuation. BenMAP-CE uses valuation functions to estimate the monetized benefits of reducing air pollution.

These benefits (reductions in risk) may vary across the population (and could be zero for some individuals). Likewise, the WTP for a given benefit is likely to vary from one individual to another. In theory, the total social value associated with the decrease in risk of a given health problem resulting from a given reduction in pollution concentrations is generally taken to be the sum of everyone's WTP for the benefits they receive.

Fundamental Concepts – Valuation

Valuation Functions are used by BenMAP-CE to estimate the economic values of changes in the incidence of health effects. In the context of human health benefits assessment, these functions help express society's preferences for avoiding certain health effects as an economic value (e.g., in U.S. dollars).

For morbidity endpoints, BenMAP-CE estimates monetized benefits by using either **Willingness to Pay (WTP)** or **Cost of Illness (COI)**-based valuation functions. **WTP** reflects the willingness of individuals to exchange money for a reduction in his or her risk of illness or death and is viewed by economists as the most complete and appropriate measure of the value of a risk reduction. **COI** estimates the value of a health effect based on the observed direct and indirect costs associated with that condition. Direct costs would include medical costs such as hospital stays and pharmaceutical costs, while indirect costs include impacts such as lost earnings from days unable to work. A COI-based estimate is expected to understate the true economic value of reductions in risk of a health effect because it does not include the value of avoided pain and suffering.

For mortality endpoints, BenMAP-CE generally estimates monetized benefits by using the **Value of Statistical Life (VSL)**, a WTP-based estimate derived from an extensive literature of observed or elicited estimates of the monetary value that an individual is willing to exchange for small reductions in his or her risk of death. It does **NOT** represent the value of the life of any one specified individual.

7.1.1.1 Monetizing Benefits

Epidemiological studies allow us to estimate the number of cases of an adverse health effect that would be avoided by a given reduction in pollutant concentrations. If we have an estimate of the average individual's WTP for the risk reduction conferred upon him, we can derive from that an estimate of the value of a statistical case avoided. Suppose, for example, that a given reduction in pollutant concentrations results in a decrease in mortality risk of 1/10,000. Then for every 10,000 individuals, one individual would be expected to die in the absence of the reduction in pollutant concentrations (who would not be expected to die in the presence of the reduction in pollutant concentrations). If the average individual's WTP for this 1/10,000 decrease in mortality risk is \$100, then the value of a statistical life is $10,000 \times \$100$, or \$1 million. In general, the ex-ante WTP for a risk reduction of x can be converted into an ex-post value of a statistical case avoided by dividing the average individual's WTP for the risk reduction of x by x (e.g. $\$100/0.0001 = \$1,000,000$). The same type of calculation can produce values for statistical incidences of other health endpoints.

Sometimes those economic values come from contingent valuation studies, in which study participants are queried about their WTP to avoid a specific adverse health effect.